# Biswadeep Chakraborty

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## Education

**Indian Institute of Science Education and Research (IISER) Bhopal** Aug 2023 – Aug 2027 (Expected)
*B.S. in Computer Science and Engineering*

**Vivekananda Kendra Vidyalaya, Nirjuli** 2020 – 2022
*Higher Secondary (CBSE): 95.3%*

**Vivekananda Kendra Vidyalaya, Nirjuli** 2010 – 2020
*Secondary (CBSE): 98%, ranked 2nd in the state*

## Research Experience

**National Chung Cheng University, Taiwan** TEEP program, Jun 2026 – Aug 2026
*Research Intern, Dept. of Computer Science & Information Engineering* *Advisor: Dr.Tsung Wei Chiang*

- Formulated heterogeneous Vision Transformer fusion as a multi-scale computational graph alignment problem, developing a three-stage Optimal Transport framework spanning macro, meso, and micro architectural structures.
- Developed an alignment pipeline combining **Fused Gromov–Wasserstein (FGW)** for block-level structural alignment, **Bures–Wasserstein barycentric synthesis** for depth-heterogeneous fusion, and entropic **Sinkhorn transport** for neuron/head-level alignment.
- Established a permutation-invariance property of Bures–Wasserstein weight synthesis, showing that hard row-axis permutation alignment is discarded during covariance-based interpolation.
- Ran controlled experiments on **CIFAR-100** and **Tiny-ImageNet** studying interpolation geometry, alignment strategy, and architectural heterogeneity; Bures–Wasserstein interpolation improved accuracy over Euclidean interpolation by up to **16.6 percentage points** on CIFAR-100.

**Indian Institute of Technology Mandi** Jun 2025 – Jul 2025
*Summer Research Intern* *Advisor: Dr. Dinesh Singh; Mentor: Tarun Saini (Ph.D.)*

- Developed and evaluated ILP, Greedy, Simulated Annealing, and a Multi-Stage algorithm for drone-based surveillance scheduling under patrol-interval constraints.
- Evaluated the algorithms on real-world maps (Ritsumeikan University and IISER Bhopal); the Multi-Stage framework achieved better coverage efficiency and lower runtime.
- Implemented **Trajectron**, a graph-based deep generative framework using CVAEs and dynamic spatiotemporal graphs for multi-agent trajectory prediction; evaluated on ETH/UCY using ADE, FDE, and NLL against baselines including SGAN.

**Indian Institute of Technology Bhubaneswar** Dec 2024 – Jan 2025
*Winter Research Intern* *Advisors: Dr. Shreya Ghosh, Dr. Abhik Jana*
Microsoft Academic Partnership Grant 2024 project: "Geospatial Knowledge Inference Using LLMs for Enhanced Travel Decision-Making."

- Developed personalised travel-planning workflows using large language models (LLMs).
- Performed web scraping, API integration, and dataset curation using Python, Playwright, and Geopy.
- Automated geospatial data processing to support scalable real-world analysis and LLM-based decision support.

## Research Projects

**Heterogeneous Transformer Fusion along Statistical Manifolds: A Bures-Wasserstein Approach** Jun 2026 – Present
*Research project from the TEEP internship at CCU, Taiwan; manuscript in preparation (first author).*

- Proposed a three-stage Optimal Transport framework that fuses Vision Transformers of different widths and depths by aligning macro (blocks), meso (heads), and micro (neurons) structure.
- Combined **Fused Gromov–Wasserstein**, **Bures–Wasserstein barycentric synthesis**, and **Sinkhorn transport**; proved that Bures–Wasserstein synthesis is invariant to hard row-axis permutation alignment.
- Benchmarked against Euclidean interpolation and other baselines on CIFAR-100 and Tiny-ImageNet, with up to 16.6 percentage points higher accuracy on CIFAR-100.

**Federated Learning for Heterogeneous Indian Judicial Data** Aug 2026 – Present
*Privacy-preserving machine learning across heterogeneous Indian judicial datasets.*

- Investigating federated learning architectures for training predictive models across decentralised judicial datasets while preserving data locality and privacy.
- Studying non-IID data distributions arising from jurisdictional differences, procedural variations, and heterogeneous case characteristics.

- Evaluating aggregation strategies for predictive performance, client-level fairness, and robustness to malicious or unreliable participants.

**Beyond the Dictionary: A Human-Centred Audio First AI Platform for Apatani Language Preservation** Oct 2026 – Present

*Independent initiative on AI-assisted learning and digital preservation of Apatani, an indigenous language of Arunachal Pradesh.*

- Conceptualising an audio-first learning platform integrating native-speaker recordings, contextual vocabulary, and conversational learning.
- Designing a community-driven dynamic dictionary linking spoken expressions with English/Hindi meanings, pronunciation guidance, and culturally grounded examples.
- Exploring multimodal retrieval and semantic search across audio, meaning, and cultural context, with community-in-the-loop validation.

**Deep Generative Systems for Indigenous Textile Archiving and Design** Jan 2026 – Present

- Designing **CNN/GAN pipelines** for regional motif classification and generative synthesis.
- Developing a searchable dataset integrating visual data, metadata, and cultural narratives.
- Prototyping an AI-assisted pattern-design toolkit and exploring neural style transfer to combine traditional motifs with contemporary aesthetics.

## Technical Skills

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| **Programming** | Python, C, SQL |
| **Machine Learning** | PyTorch, Scikit-learn, OpenCV |
| **Data & Visualisation** | Pandas, NumPy, Matplotlib, Seaborn, NetworkX |
| **Mathematics** | Linear Algebra, Probability, Statistics, Optimization, Calculus |
| **Web & Automation** | Playwright, Geopy, API Integration |
| **Tools** | Jupyter Notebook, Mathematica, AutoCAD, Microsoft Office |

## Awards & Achievements

- **Acceptance to New York University**, Computer Science & Mathematics, Class of 2026 2022
- **Dera Natung Award for Academic Excellence**, Government of Arunachal Pradesh 2021
- **State Merit Scholarship**, ranked 1$^{st}$ in the state, Government of Arunachal Pradesh 2019

## Leadership & Extracurricular Activities

**NEVISTA, North-East Fest of IISER Bhopal** Jan 2026

*Co-Founder and Convener*

Co-founded and served as the first Convener of a cultural festival celebrating the heritage and diversity of North-East India.

**IISER Bhopal Quiz Club** Dec 2024 – Present

*Core Committee Member*

Quiz hosting, content development, and event organisation.

**The Poetry Club, IISER Bhopal** Sep 2024 – Apr 2025

*Core Committee Member*

Created and edited promotional videos and managed digital media for club events.

**Animal Welfare Society, IISER Bhopal** Feb 2024 – Aug 2024

*Content Creation Team*

Produced content and media supporting animal welfare initiatives on campus.

**Be The Voice** Jul 2020 – Dec 2021

*Co-Founder*

Co-founded an NGO for stray animal feeding, medical treatment, adoption, and abuse reporting in the Itanagar capital region.

## Relevant Coursework

Machine Learning (ECS317), Applied Optimization (ECS311), Probability and Statistics (MTH202), Linear Algebra (MTH102), Multivariable Calculus (MTH201), Discrete Mathematics I (ECS201), Data Structures and Algorithms (ECS202), Theory of Computation (ECS307), Fundamentals of Database Systems (ECS312), Introduction to Programming (ECS102),